

Plasma Cell — War Room (Diagnosis, ISS Staging & Genetics)



1. CD138 plasma cells greater than or equal to 10%

WHAT YOU SEE

PROOF OF INVASION microscope, CD138 PLASMA CELLS greater than or equal to 10%

WHAT IT ENCODES

Diagnosis of multiple myeloma requires clonal bone-marrow plasma cells greater than or equal to 10% OR a biopsy-proven plasmacytoma (bony or extramedullary). CD138 (syndecan-1) is the immunohistochemical marker used to identify and quantify plasma cells, and clonality is confirmed by kappa/lambda light-chain restriction (flow or in-situ hybridization). Mnemonic: CD138 is the 'PROOF OF INVASION' badge under the microscope.

BOARD TRAP

WRONG answer: 'greater than or equal to 10% clonal plasma cells alone diagnoses multiple myeloma.' By itself this only establishes a plasma-cell neoplasm — you ALSO need a myeloma-defining event (CRAB end-organ damage OR a SLiM biomarker). Discriminator: greater than or equal to 10% plasma cells WITHOUT CRAB/SLiM = smoldering myeloma; <10% with small M-protein and no events = MGUS.

2. CRAB report card

WHAT YOU SEE

EVIDENCE OF DAMAGE CRAB REPORT CARD

WHAT IT ENCODES

CRAB = the end-organ damage that defines symptomatic myeloma: hyperCalcemia (>11 mg/dL or >1 above ULN), Renal insufficiency (Cr >2 mg/dL or CrCl <40), Anemia (Hgb <10 or >2 g/dL below normal), and Bone lesions (≥1 lytic lesion on skeletal survey/CT/PET). Any one CRAB feature attributable to the clone establishes active myeloma needing treatment. Mnemonic: the CRAB report card grades the four organs.

BOARD TRAP

WRONG answer: 'a single lytic lesion isn't enough — you need multiple CRAB features to call it myeloma.' Just ONE CRAB criterion (or one SLiM biomarker) attributable to the plasma-cell clone is sufficient. Discriminator: the renal lesion is classically cast nephropathy from free light chains (not a simple prerenal azotemia), and bone disease is LYTIC (contrast POEMS = sclerotic).

3. SLiM criteria

WHAT YOU SEE

SLIM INTELLIGENCE: greater than or equal to 60% plasma cells, greater than or equal to 100 FLC ratio, greater than 1 MRI lesion

WHAT IT ENCODES

SLiM biomarkers are the 2014 IMWG additions that define myeloma even WITHOUT CRAB, because they predict near-certain progression: Sixty% — clonal marrow plasma cells greater than or equal to 60%; Light chains — involved:uninvolved serum free light-chain ratio greater than or equal to 100 (with involved FLC ≥100 mg/L); MRI — greater than 1 focal lesion (each ≥5 mm) on MRI. Mnemonic: SLiM = Sixty, Light-chain 100, MRI >1.

BOARD TRAP

WRONG answer: 'a free light-chain ratio of 8 (abnormal) meets the SLiM criterion and diagnoses myeloma.' The SLiM cutoff is a ratio greater than or equal to 100 (with involved FLC ≥100 mg/L) — modestly abnormal ratios do not qualify. Discriminator: an FLC ratio of 8 with no CRAB and 15% plasma cells is smoldering myeloma, not active myeloma.

4. ISS Rank I — B2M less than 3.5, albumin greater than or equal to 3.5

WHAT YOU SEE

ISS STAGING BOARD RANK I, GAS PEDAL B2-microglobulin less than 3.5, albumin

WHAT IT ENCODES

ISS Stage I requires BOTH beta-2-microglobulin <3.5 mg/L AND albumin greater than or equal to 3.5 g/dL — the best-prognosis group (median OS historically ~62 months in original ISS). It reflects low tumor burden with preserved nutritional/hepatic synthetic status. Mnemonic: Rank I = low gas pedal (B2M) AND high albumin = the longest survival.

BOARD TRAP

WRONG answer: 'a low B2-microglobulin alone puts the patient in ISS Stage I.' Stage I needs BOTH a low B2M AND a normal-or-high albumin — a low B2M with low albumin is Stage II. Discriminator: ISS is defined by the combination of the two labs, not B2M in isolation.

5. ISS Rank II — intermediate

WHAT YOU SEE

RANK II medal on staging board

WHAT IT ENCODES

ISS Stage II is defined by exclusion: NOT Stage I and NOT Stage III. In practice this is either B2M <3.5 with albumin <3.5, OR B2M 3.5–5.5 (any albumin). It is the intermediate-prognosis group (original median OS ~44 months). Mnemonic: Rank II = the middle medal, 'neither best nor worst.'

BOARD TRAP

WRONG answer: 'ISS Stage II has its own specific lab thresholds you must memorize.' Stage II is purely a residual category — anything that is not Stage I and not Stage III. Discriminator: anchor on the Stage I criteria (B2M <3.5 AND albumin ≥3.5) and the Stage III criterion (B2M >5.5); everything between is Stage II.

6. ISS Rank III — B2M greater than 5.5

WHAT YOU SEE

RANK III, BRAKES, B2M greater than 5.5

WHAT IT ENCODES

ISS Stage III is beta-2-microglobulin >5.5 mg/L (regardless of albumin) — the highest tumor burden / worst prognosis (original median OS ~29 months). High B2M reflects both large tumor mass and renal impairment. Mnemonic: Rank III = B2M floored past 5.5 = brakes on survival.

BOARD TRAP

WRONG answer: 'ISS staging incorporates LDH and cytogenetics.' The original ISS uses ONLY B2-microglobulin and albumin. LDH and high-risk FISH cytogenetics are added in the REVISED R-ISS, which reclassifies prognosis (R-ISS I/II/III). Discriminator: if a stem mentions LDH or del(17p)/t(4;14) affecting stage, it is testing R-ISS, not ISS.

7. Beta2-microglobulin = gas pedal

WHAT YOU SEE

GAS PEDAL labeled B2-microglobulin

WHAT IT ENCODES

Beta-2-microglobulin is the light chain of MHC class I shed by all nucleated cells; in myeloma it rises with tumor BURDEN and with renal impairment (it is renally cleared), so elevated B2M can reflect either more disease or worse kidneys. It is the dominant adverse prognostic lab in ISS. Mnemonic: B2M is the GAS PEDAL — the higher it goes, the faster toward poor prognosis.

BOARD TRAP

WRONG answer: 'an elevated B2-microglobulin always means high tumor burden.' Because B2M is renally cleared, renal failure (even from cast nephropathy) elevates it independently of tumor mass. Discriminator: interpret B2M alongside creatinine — and remember it is a prognostic, not diagnostic, marker.

8. Albumin = brakes

WHAT YOU SEE

BRAKES labeled ALBUMIN

WHAT IT ENCODES

Serum albumin is INVERSELY prognostic: low albumin (driven partly by IL-6, the plasma-cell growth factor, and by poor nutrition) worsens stage, whereas higher albumin is protective. In ISS, albumin ≥ 3.5 g/dL is the favorable cutoff. Mnemonic: albumin is the BRAKES — more albumin slows progression and improves prognosis.

BOARD TRAP

WRONG answer: 'higher albumin worsens prognosis / albumin and B2M move the stage in the same direction.' They move OPPOSITELY — high B2M is bad, high albumin is GOOD. Discriminator: Stage I needs the favorable pairing (low B2M + high albumin); mixing them up flips the staging.

9. High-risk genetics

WHAT YOU SEE

HIGH-RISK GENETICS WALL del(17p), t(4;14), t(14;16)

WHAT IT ENCODES

High-risk cytogenetics by FISH: del(17p) (loss of TP53), t(4;14) (FGFR3/MMSET), and t(14;16) (MAF) — also t(14;20) and amp/gain(1q), del(1p). These confer poor prognosis and contribute to R-ISS Stage III. They influence treatment intensity (e.g., favoring carfilzomib-based regimens and consideration of tandem ASCT). Mnemonic: the High-Risk wall mugshots = del(17p), t(4;14), t(14;16).

BOARD TRAP

WRONG answer: 't(11;14) is a high-risk translocation requiring intensified therapy.' t(11;14) is STANDARD/favorable risk (and notably predicts BCL2-dependence → venetoclax sensitivity). Discriminator: the high-risk IgH translocations are t(4;14) and t(14;16); del(17p) is the key high-risk deletion. Don't lump all 14q translocations together.

10. Standard-risk genetics

WHAT YOU SEE

STANDARD-RISK GENETICS, hyperdiploid

WHAT IT ENCODES

Standard/favorable-risk genetics: HYPERDIPLOIDY (trisomies of odd-numbered chromosomes) and t(11;14). These carry better prognosis with current therapy. Notably t(11;14) marks BCL2 dependence, predicting response to venetoclax — a board-relevant targeted option. Mnemonic: the Standard-Risk wall shows the 'extra-chromosome' hyperdiploid clone.

BOARD TRAP

WRONG answer: 'hyperdiploidy indicates aggressive, high-risk myeloma.' Hyperdiploidy is a STANDARD/favorable-risk feature. Discriminator: aggressive biology comes from del(17p), t(4;14), t(14;16), and 1q gain — not from hyperdiploidy or t(11;14).

11. FISH vs karyotype

WHAT YOU SEE

FISH PROBE vs KARYOTYPE, FISH WORKS

WHAT IT ENCODES

Myeloma plasma cells have a LOW proliferative index, so they rarely yield metaphases for conventional karyotyping. Interphase FISH (often on CD138-selected/sorted plasma cells) detects the diagnostically and prognostically important abnormalities — del(17p), t(4;14), t(14;16), 1q gain — and is the standard cytogenetic test. Mnemonic: the FISH probe lights up where the karyotype comes up empty.

BOARD TRAP

WRONG answer: 'order a standard metaphase karyotype to risk-stratify the myeloma.' Conventional karyotype is LOW-YIELD because plasma cells seldom divide in culture — interphase FISH (ideally on plasma-cell-enriched samples) is the correct test. Discriminator: a 'normal karyotype' does NOT exclude high-risk disease; you must do FISH.

12. Key serum labs

WHAT YOU SEE

LABORATORY TOOLKIT: SPEP, immunofixation, FLC, CBC, ANION GAP DECREASED, DIPSTICK NEGATIVE

WHAT IT ENCODES

Core workup: SPEP (quantifies the M-spike), serum immunofixation (identifies the heavy/light-chain isotype), serum free light chains (essential for light-chain/oligosecretory and nonsecretory disease), 24-hour UPEP/urine immunofixation (Bence Jones), plus CBC, calcium, creatinine, and a skeletal survey/whole-body low-dose CT or PET. Cationic IgG can NARROW the anion gap. Mnemonic: SPEP + IFE + FLC + UPEP is the diagnostic toolkit.

BOARD TRAP

WRONG answer: 'a positive urine DIPSTICK for protein detects Bence Jones protein and screens for myeloma kidney.' The urine dipstick detects mainly ALBUMIN and is typically NEGATIVE despite massive light-chain (Bence Jones) proteinuria — you need UPEP/urine immunofixation or a urine protein:creatinine ratio. Discriminator: a negative dipstick with heavy proteinuria on quantitative testing = light-chain (overflow) proteinuria of myeloma.

13. IgG most common isotype

WHAT YOU SEE

KEY SERUM FACTS IgG 53% most common, IgA, ALP

WHAT IT ENCODES

IgG is the most common myeloma isotype (~50–60%), followed by IgA, then light-chain-only; IgD and IgM myeloma are rare. Alkaline phosphatase (ALP) is characteristically NORMAL despite extensive lytic bone disease because osteoBLAST activity is suppressed by the clone (DKK1, etc.). Mnemonic: IgG leads at ~53%; ALP stays flat even as bones break down.

BOARD TRAP

WRONG answer: 'lytic bone lesions should raise the alkaline phosphatase, so a normal ALP argues against bone involvement.' In myeloma ALP is usually NORMAL because osteoblasts are inhibited (uncoupled bone turnover) — unlike osteoblastic metastases or Paget disease where ALP rises. Discriminator: normal ALP with painful lytic lesions and a monoclonal protein is classic for myeloma.

14. Morphology corner

WHAT YOU SEE

MORPHOLOGY CORNER eccentric nucleus, clock-face chromatin, perinuclear hof

WHAT IT ENCODES

Plasma-cell morphology: ECCENTRIC nucleus, 'clock-face' / spoke-wheel (cartwheel) heterochromatin, abundant basophilic cytoplasm, and a perinuclear HOF (pale paranuclear zone = the prominent Golgi). On the peripheral smear, the high paraprotein causes ROULEAUX (stacked-coin RBCs). Mnemonic: clock-face nucleus + perinuclear hof = the textbook plasma cell.

BOARD TRAP

WRONG answer: 'rouleaux on the smear is specific for myeloma.' Rouleaux merely reflects a high plasma-protein/paraprotein state and is seen in any hyperglobulinemia/inflammation — it is suggestive, not diagnostic. Discriminator: confirm with SPEP/IFE/FLC and marrow; clock-face chromatin + perinuclear hof identify the plasma cell itself, but clonality (light-chain restriction) is what proves neoplasia.